Report: Document Classification using NLP Techniques

Executive Summary

This report presents a Natural Language Processing (NLP)-based solution for automating document classification. Using a dataset of 2,225 BBC news articles, we developed a system that classifies documents into five categories with 96% accuracy. The project demonstrates how NLP and machine learning can significantly reduce manual effort, improve classification accuracy, and scale document handling operations. The solution leverages Python, TF-IDF vectorization, and Logistic Regression to deliver high performance with minimal computational overhead.

Business Context & Problem Statement

Organizations today manage vast volumes of unstructured text data, from emails and reports to customer feedback and news articles. Manually categorizing these documents is time-consuming, inconsistent, and prone to human error. Automated document classification using NLP offers a scalable solution, enabling faster decision-making, improved customer service, and reduced operational costs. This project addresses the need for a reliable, automated classification system that can be integrated into existing workflows.

Approach & Methodology

We adopted a structured NLP pipeline to build the classification system. The process included:

- Data Collection: Using a labeled dataset of BBC news articles.
- Preprocessing: Cleaning and preparing text by removing noise and standardizing format.
- Feature Extraction: Converting text into numerical form using TF-IDF.
- Model Training: Applying Logistic Regression to learn from the data.
- Evaluation: Measuring performance using accuracy, precision, recall, and F1-score.

- ## Implementation Highlights

The system was implemented in Python using the following tools:
- Pandas: Data manipulation and analysis.
- NLTK: Text preprocessing (stopword removal, tokenization).
- Scikit-learn: TF-IDF vectorization and Logistic Regression.
- Matplotlib & Seaborn: Visualization of results.

The modular codebase ensured efficient development and easy maintenance. The entire pipeline was executed with minimal computational resources, making it suitable for deployment in business environments.

- ## 5. Results & Business Impact

The model achieved 96% accuracy on unseen test data, correctly classifying 427 out of 445 documents. High precision and recall across all categories indicate reliable performance. This translates to significant business benefits:
- Time Savings: Automates classification of thousands of documents in seconds.
- Error Reduction: Minimizes human error in categorization.
- Scalability: Easily adapts to larger datasets and new categories.

- ## Insights & Challenges

Key insights from the project include:
- Traditional models like Logistic Regression can deliver high accuracy when paired with effective preprocessing.
- TF-IDF is a powerful feature extraction method for text classification.

Challenges faced:
- Semantic Overlap: Categories like Business and Politics had overlapping vocabulary, leading to some misclassifications.
- Scalability: While effective on the current dataset, larger datasets may require more advanced models or infrastructure.

- ## Recommendations

To enhance the system and maximize business value, we recommend:
- Integration: Embed the classifier into document management systems for real-time categorization.

- Advanced Models: Explore deep learning models like BERT for improved semantic understanding.
- Dataset Expansion: Train on larger, more diverse datasets to improve generalization.
- Monitoring: Implement feedback loops to continuously improve model accuracy.

• Conclusion

This project demonstrates that NLP-based document classification can transform manual, error-prone processes into efficient, automated workflows. With 96% accuracy, the system is ready for real-world deployment and offers a foundation for further innovation using advanced AI models. Adopting such solutions can lead to measurable improvements in productivity, accuracy, and scalability.

• Appendix: Technical Details

- Dataset: BBC Text Dataset with 2,225 labeled news articles.
- Preprocessing: Lowercasing, punctuation removal, stopword filtering, tokenization.
- Feature Extraction: TF-IDF vectorization.
- Model: Logistic Regression with 80/20 train-test split.
- Metrics: Accuracy (96%), Precision, Recall, F1-score.
- Tools: Python, Pandas, NLTK, Scikit-learn, Matplotlib, Seabor